

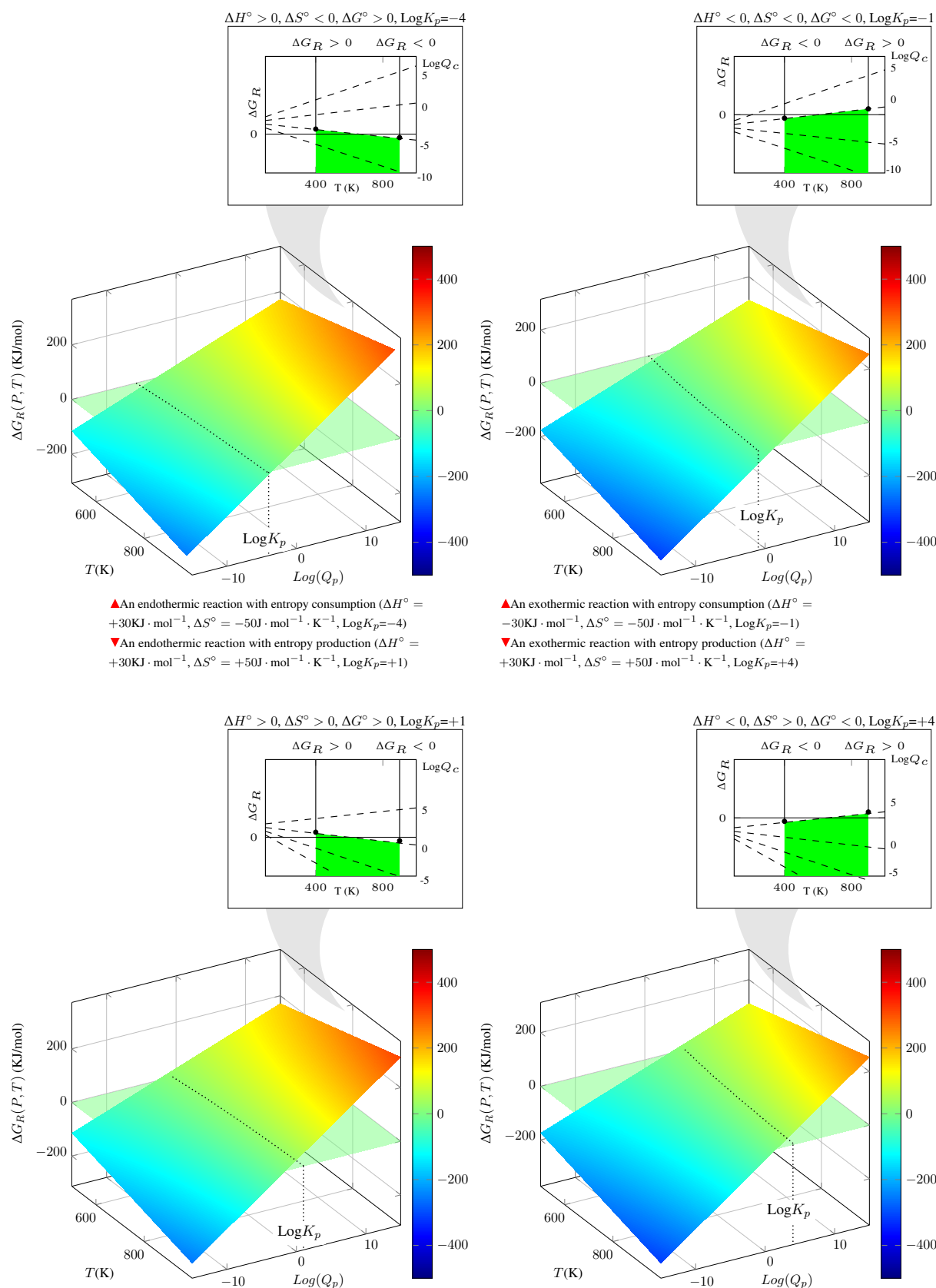


Figure 1 Gibbs free energy 3D plot for a set of reaction with different internal thermodynamics parameters: (Top Left) an endergonic reaction: you can increase T to favour the reaction only for pressure conditions ($\text{Log} Q_c$) lower than $\Delta S/R$; (Top Right) an exergonic reaction mediated by H : you disfavour the reaction by increasing T only for pressure conditions ($\text{Log} Q_c$) higher than $\Delta S/R$; (Bottom Left) an endergonic reaction mediated by S : you can increase T to favour the reaction only for pressure conditions ($\text{Log} Q_c$) lower than $\Delta S/R$; (Bottom Right) an exergonic reaction mediated by both the H and S : you disfavour the reaction by increasing T only for pressure conditions ($\text{Log} Q_c$) higher than $\Delta S/R$;

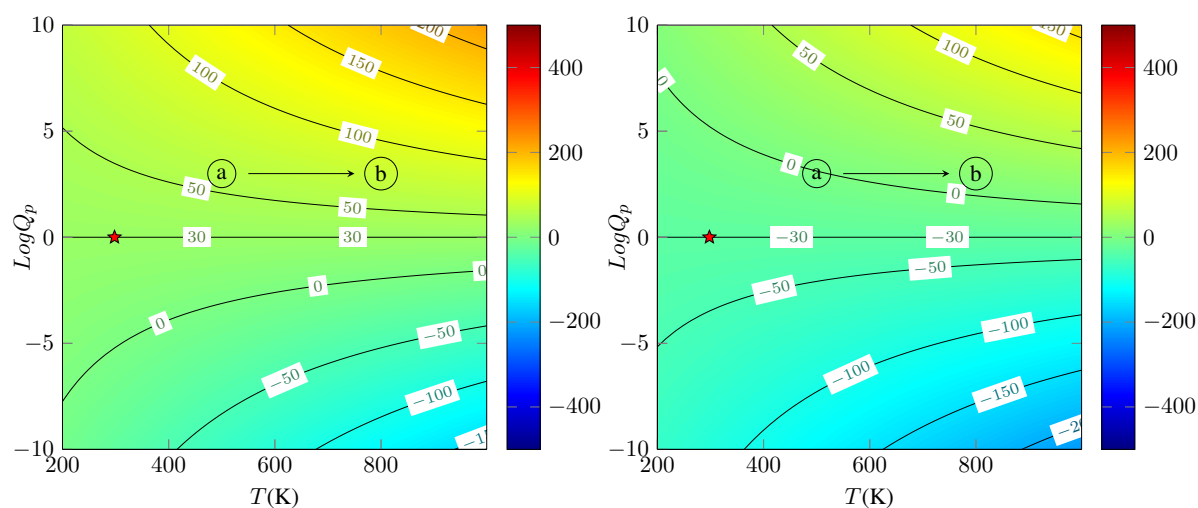


Figure 2 Gibbs free energy 3D plot displaying contour lines for different pressure and temperature conditions for an endergonic (Left Panel) and an exergonic (Right Panel) reaction. The star sign indicated the standard pressure and temperature conditions. The reaction on the left is non-spontaneous under the conditions labeled as *a* on the left panel ($\Delta G_R > 0$), whereas the reaction on the right is spontaneous under the same conditions ($\Delta G_R < 0$). Increasing the temperature until reaching the conditions labeled as *b* does not affect the spontaneity of the reaction on the left, whereas it impedes the reaction on the right, which is now non-spontaneous under *b* conditions ($\Delta G_R > 0$).